

The 65th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Intestinal Geographical Pattern of Gut Microbiome and Metabolites Identifies Novel Regulators of Graft-Versus-Host Disease*Emma Lauder¹, Anders Kiledal, PhD², Dongchang Zhao, PhD³, Gregory Dick, PhD⁴, Pavan Reddy, MD⁵*¹ University of Michigan, Ann Arbor, MI² Department of Earth and Environmental Sciences, University of Michigan, Ann Arbor, MI³ Baylor College of Medicine, Houston, TX⁴ Rogel Cancer Center, University of Michigan, Ann Arbor, MI⁵ Dan L Duncan Comprehensive Cancer Center, Baylor College of Medicine, Houston

Introduction: The severity of T cell-mediated gastrointestinal graft-versus-host disease (GI-GVHD) correlates with a decrease in the gut microbiome diversity, loss of obligate anaerobic bacteria and alterations in levels of salutary metabolites like short chain fatty acids (SCFAs). However, it is unknown whether the intestinal anatomic geography of the microbiome and its metabolites impacts GI GVHD severity. Herein, we investigate the pattern of the gut microbiome and metabolites from four distinct anatomical sites of intestines after allogeneic stem cell transplant (allo-SCT) via shotgun metagenomics and untargeted mass-spectrometry.

Methods: To determine the composition of the gut microbiome and gut metabolites of mice after allo-SCT, we utilized contemporaneous analyses of shotgun metagenomic sequencing and untargeted mass spectrometry on the gut contents of mice 7 or 21 days after major histocompatibility complex (MHC)-disparate BALB/c→C57BL/6 (B6) allogeneic SCT. We investigated the microbial community composition and gut metabolites from four distinct gut locations (terminal ileum, cecum, transverse colon, descending colon) and performed integrative analysis to assess for novel pathways that may regulate GVHD biology. For shotgun metagenomics analyses, reads were depth normalized with BBnorm v.38.96 and assembled per-sample with MEGAHIT v.1.2.9. Prodigal v.2.6.3 was used to find genes in each bin. KofamScan v.1.3.0 was used to assign KEGG ortholog IDs (KO) to translated genes predicted by Prodigal. Community composition profiles were produced with Kraken2 v.2.1.2 and Bracken v.2.6.1. Functional profiles for each sample were determined with the Kraken community composition profiles and Humann3 v.3.0.0, with a GTDB database built with Struo2.

Results: Consistent with previous reports from direct stool analyses, our data demonstrated that after allo-SCT, gut microbial diversity is reduced, relative abundance of obligate anaerobes decreases, and relative abundance of facultative anaerobes increases at all the intestinal anatomic sites, but the difference was maximal in ileo-cecal contents. Furthermore, metabolomic analyses reveal a decrease in previously described metabolites such as, SCFAs, bile acids and indoles at these sites. The distinct patterns of change between control and allogeneic samples depended not only on the gut location probed but also correlated with timepoint after transplant (7 days or 21 days) highlighting the importance of probing different gut locations and timepoints. The largest driver of differences between sample groups was (1) gut location (2) time after transplant (3) transplant type.

Integrative analyses demonstrated loss of microbial functional genes related to butyrate processing following allo-SCT complementing previous studies that demonstrated reduction in butyrate levels. Further bioinformatic analyses identified novel microbial functional genes and metabolites that may have biological implications. Analyses revealed that after allo-SCT, phenyllactic acid (PLA) levels increased in the transverse colon and distal colon (fig.1). PLA is an anti-microbial compound and has been shown to have immune modulatory properties. Administration of PLA in drinking water of mice after allo-SCT worsened GVHD severity and increased mortality indicating PLA may play a role in inducing dysbiosis and worsening GVHD severity.

Conclusion: Contemporaneous analyses of shotgun metagenomic and untargeted mass spectrometry of gut contents after allo-SCT reveal changes in microbial composition and metabolites consistent with previous data from analyses from stool. However, our analysis demonstrates distinct patterns of key metabolites and microbes in between syngeneic and allogeneic depending on gut location and time after transplant. Furthermore, our analyses have identified novel functional genes and metabolites that are implicated in GVHD severity.

Disclosures No relevant conflicts of interest to declare.

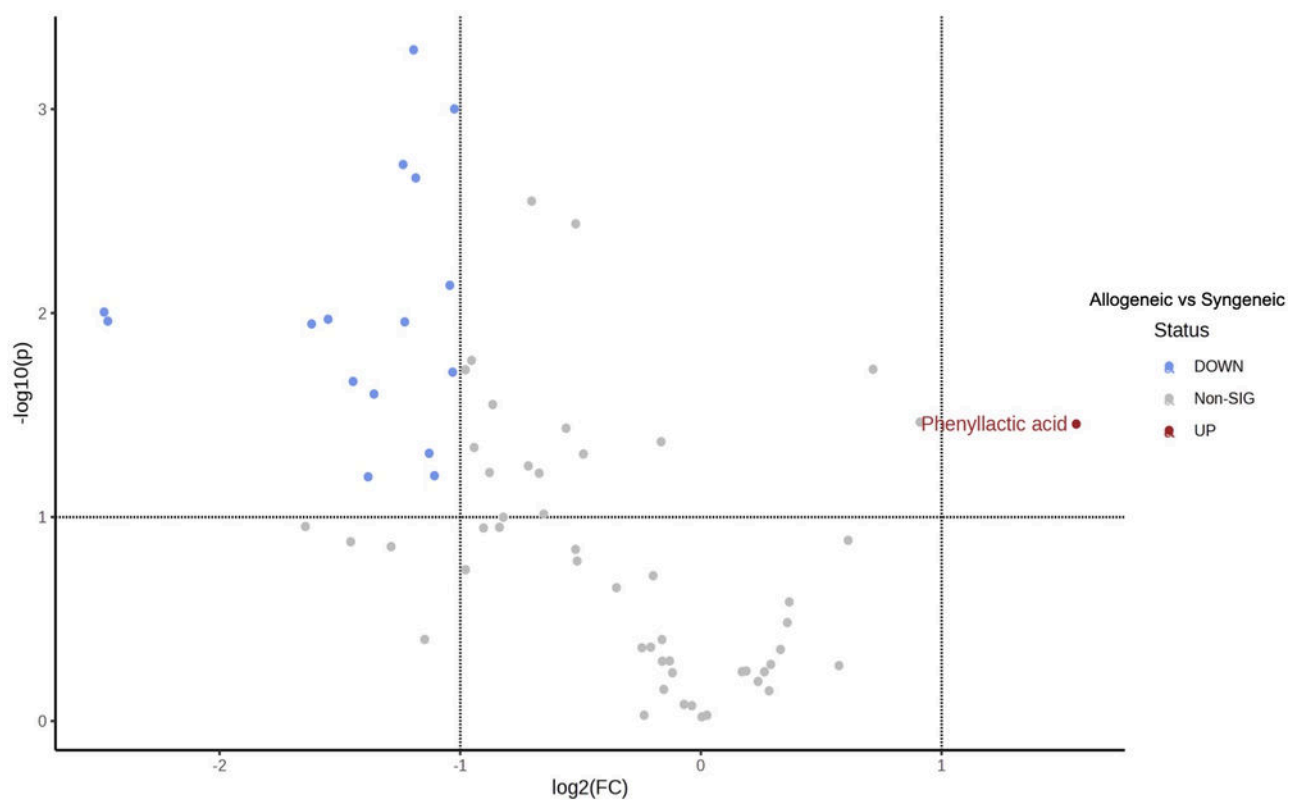


Figure 1. Differentially abundant metabolites in the Descending Colon on Day 21 after Transplant

Figure 1

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